

TABLE II

Examples of Character Formation on a 7-Segment Display Module

Character		dp	g	f	e	d	c	b	a
0	0b	1	1	0	0	0	0	0	0
1	0b	1	1	1	1	1	0	0	1
h	0b	1	0	0	0	1	0	1	1
d	0b	1	0	1	0	0	0	0	1
y	0b	1	0	0	1	0	0	0	1
l	0b	1	1	0	0	0	1	1	1
.	0b	0	1	1	1	1	1	1	1
p	0b	0	0	0	0	1	1	0	0
Blank	0b	1	1	1	1	1	1	1	1

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com

using the Arduino board. After uploading, the chip is put on the target board. For the slave clock circuit, Arduino Nano is used.

Table II explains how characters are formed on a 7-segment display module. For a common-anode type 7-segment display, 1 means off and 0 means on. All characters are dumped in a digit [] array and their sequence is dumped in a seq [] array.

```
int seq[14] = {1,2,4,8,16,32,64,128}; //each byte represents the digit from right to left.
```

```
shiftOut(datapin,clockpin,MSBFI  
RST,seq[pos-1]); //0,1,2,3,4,...7
```

```
shiftOut(datapin,clockpin,MSBFI  
RST,digits[number]); //number
```

The first byte will go to the shift register responsible for the position digit, while the second byte will write the character on the 7-segment display. Every segment will glow sequentially, but since the speed at which these are operating is very fast, we can see these as readable characters in a meaningful way.

The speed information will not show on the display unless you are on the move. So, put the master clock in a car and drive along; you will see the speed on the display.

The normal range of slave clocks from the master clock is about 100 metres. ●



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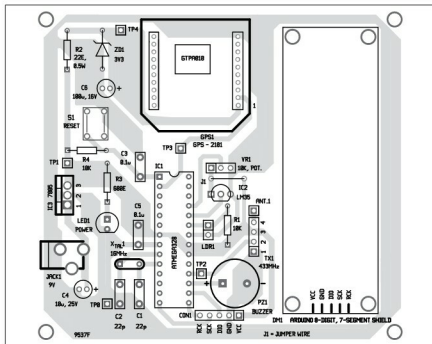


Fig. 7: Component layout of the PCB of the GPS master clock

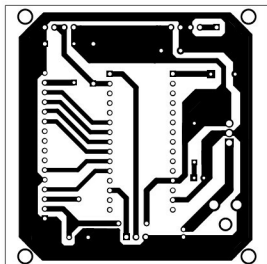


Fig. 8: Actual-size PCB layout of the GPS slave clock

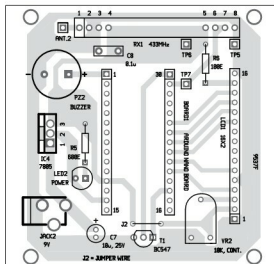


Fig. 9: Component layout of the PCB of the GPS slave clock